Aakash
Medical | IIT-JEE | Foundations
Corporate Office: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

FINAL TEST SERIES for NEET-2024

MM : 720

Test - 8

Time : 3 Hrs. 20 Mins.

Answers

1. (2)	41. (3)	81. (1)	121. (2)	161. (4)
2. (2)	42. (3)	82. (2)	122. (3)	162. (3)
3. (2)	43. (4)	83. (1)	123. (4)	163. (1)
4. (2)	44. (1)	84. (1)	124. (1)	164. (2)
5. (2)	45. (1)	85. (1)	125. (2)	165. (4)
6. (1)	46. (3)	86. (4)	126. (3)	166. (1)
7. (3)	47. (2)	87. (1)	127. (3)	167. (3)
8. (1)	48. (1)	88. (1)	128. (4)	168. (4)
9. (2)	49. (4)	89. (2)	129. (3)	169. (3)
10. (4)	50. (2)	90. (2)	130. (2)	170. (3)
11. (3)	51. (2)	91. (4)	131. (4)	171. (4)
12. (4)	52. (3)	92. (2)	132. (3)	172. (3)
13. (4)	53. (3)	93. (1)	133. (1)	173. (2)
14. (1)	54. (3)	94. (1)	134. (2)	174. (4)
15. (2)	55. (2)	95. (4)	135. (4)	175. (4)
16. (2)	56. (4)	96. (1)	136. (1)	176. (3)
17. (3)	57. (2)	97. (3)	137. (3)	177. (2)
18. (3)	58. (2)	98. (2)	138. (1)	178. (4)
19. (4)	59. (2)	99. (1)	139. (4)	179. (4)
20. (3)	60. (1)	100. (4)	140. (3)	180. (3)
21. (2)	61. (1)	101. (2)	141. (1)	181. (1)
22. (2)	62. (2)	102. (3)	142. (3)	182. (2)
23. (2)	63. (4)	103. (4)	143. (2)	183. (3)
24. (3)	64. (1)	104. (1)	144. (1)	184. (4)
25. (2)	65. (2)	105. (3)	145. (2)	185. (2)
26. (4)	66. (2)	106. (4)	146. (1)	186. (2)
27. (2)	67. (4)	107. (2)	147. (4)	187. (3)
28. (3)	68. (2)	108. (4)	148. (3)	188. (2)
29. (4)	69. (3)	109. (3)	149. (1)	189. (2)
30. (4)	70. (3)	110. (3)	150. (4)	190. (3)
31. (3)	71. (4)	111. (2)	151. (3)	191. (3)
32. (3)	72. (4)	112. (3)	152. (2)	192. (4)
33. (2)	73. (3)	113. (4)	153. (4)	193. (1)
34. (2)	74. (2)	114. (2)	154. (1)	194. (1)
35. (1)	75. (4)	115. (2)	155. (2)	195. (1)
36. (4)	76. (2)	116. (1)	156. (4)	196. (2)
37. (2)	77. (1)	117. (4)	157. (3)	197. (4)
38. (1)	78. (2)	118. (2)	158. (3)	198. (2)
39. (2)	79. (4)	119. (1)	159. (2)	199. (2)
40. (2)	80. (4)	120. (2)	160. (3)	200. (2)

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Answers and Solutions

PHYSICS

SECTION - A

1. Answer (2)

Minimum 2 NOR gates are required to form an OR gate.

2. Answer (2)

A	B	Y
1	1	1
0	0	1

3. Answer (2)

$$E = \frac{hc}{\lambda}$$

$$= \frac{12400}{\lambda(\text{Å})}$$

$$= \frac{12400}{6200}$$

= 2 eV

4. Answer (2)

$$E = \frac{hc}{\lambda}$$

If λ is doubled, then energy become halved.

5. Answer (2)

de-Broglie relation $\rightarrow \lambda = \frac{h}{p}$

Threshold frequency $\rightarrow \nu = \frac{\phi_0}{h}$

Einstein photoelectric equation $\rightarrow K_{\max} = h\nu - \phi_0$

6. Answer (1)

Electron volt is the unit of energy.

7. Answer (3)

Coulomb force is responsible for α-particle scattering.

8. Answer (1)

P.E. = 2 × T.E.

= 2 × (–13.6)

= –27.2 eV

9. Answer (2)

$E = \Delta mc^2$

$= 5 \times 10^{-3} \times 10^{-3} \times (3 \times 10^8)^2$

$= 5 \times 10^{-6} \times 9 \times 10^{16}$

$= 45 \times 10^{10}$ J

10. Answer (4)

Radius of nucleus $R = R_0(A)^{\frac{1}{3}}$

$$\frac{R_{\text{Li}}}{R_{\text{Fe}}} = \left(\frac{7}{56}\right)^{\frac{1}{3}}$$

$$= \left(\frac{1}{8}\right)^{\frac{1}{3}} = \frac{1}{2}$$

11. Answer (3)

Any semiconductor material whether N-type or P-type is electrically neutral.

12. Answer (4)

The current through the diode is zero since it is reverse biased.

$\therefore I = 0$

13. Answer (4)

$\tau_{restoring} = c\theta$

$\Rightarrow NiAB = c\theta$

$\frac{\theta}{i} = \frac{NAB}{c}$

14. Answer (1)

$\lambda = \frac{h}{P}$

$\Rightarrow P_P = P_D$

$\Rightarrow m_P v_P = m_D v_D$

$\Rightarrow m v_P = 2m v_D$

$\frac{v_P}{v_D} = 2$

15. Answer (2)

Vernier constant or Least count is given by

V.C. = 1MSD – 1 VSD

16. Answer (2)

End correction $e = \frac{l_2 - 3l_1}{2}$

$= \frac{80 - 3 \times 25}{2}$

$= \frac{5}{2}$ = 2.5 cm

17. Answer (3)

Higher the stopping potential greater will be incident radiation frequency. V_{03} is highest stopping potential so ν_3 is highest frequency.

18. Answer (3)

$\lambda = \frac{h}{\sqrt{2mqV}}$

$\therefore \quad \lambda = \frac{h}{\sqrt{2mq}} \times \frac{1}{\sqrt{V}}$

$\therefore$ Slope $\propto \frac{1}{\sqrt{m}}$

$\therefore \quad m_A > m_C > m_B$

19. Answer (4)

$K_{max} = eV_0 = h\nu - h\nu_0 = h\nu - \phi_0$

ν is maximum for violet colour.

20. Answer (3)

Photo-electric cell convert light energy into electric energy.

21. Answer (2)

Current $\propto$ intensity (given frequency)

Stopping potential $\propto$ frequency (given intensity)

22. Answer (2)

de-Broglie wavelength associated with any electron.

$\lambda = \frac{12.27}{\sqrt{V}}$ Å

$\lambda = \frac{12.27}{\sqrt{64}} = \frac{12.27}{8} = 1.53$ Å

23. Answer (2)

$E_g \le \frac{hc}{\lambda}$

$\lambda \le \left[\frac{12400}{E_g\,(\text{eV})}\right]$ Å

$\lambda \le \frac{12400}{2.2}$

$\lambda \le 5636$ Å

24. Answer (3)

I-*V* characteristic of photodiode lies in 3rd quadrant of co-ordinate axis while of solar cell lies in 4th quadrant.

25. Answer (2)

In p-type semiconductor fermi level shifts towards valence band and in n-type it shifts towards conduction band.

26. Answer (4)

$r_n \propto n^2$

$\frac{R}{R_1} = \left(\frac{2}{3}\right)^2$

$R_1 = \frac{9R}{4}$

27. Answer (2)

Isotopes have same number of protons i.e. $Z_1 = Z_2$

Isobars have same number of nucleons i.e. $A_1 = A_2$

Isotones have same number of neutrons i.e. $n_1 = n_2$

or $A_1 - Z_1 = A_2 - Z_2$

28. Answer (3)

Nuclear force is charge independent.

29. Answer (4)

$\Delta m = (3m_p + 4m_n - m_{Li})$

$= [3(1.007277) + 4(1.008665) - 7.016005]$

= 0.040486 amu

30. Answer (4)

$n \longrightarrow p + e^- + \bar{\nu}$

$p \longrightarrow n + e^+ + \nu$

Free proton is stable while free neutron is unstable.

${}_Z X^A \xrightarrow{\beta \text{ decay}} {}_{Z\pm1}Y^A$

31. Answer (3)

$n_e n_h = n_i^2$

$n_e = \frac{(1.5\times10^{10})^2}{4.5\times10^{16}} = 5\times10^3 \text{ cm}^{-3}$

$n_e = 5\times10^3\times10^6 \text{ m}^{-3}$

$= 5 \times 10^9 \text{ m}^{-3}$

32. Answer (3)

${}^{m}_{n}X \xrightarrow{-2\alpha} {}^{m-8}_{n-4}Y_1 \xrightarrow{-2\beta^-} {}^{m-8}_{n-2}Y$

33. Answer (2)

$Q = BE_{product} - BE_{Reactant}$

$= E_2 - E_1 - E_1$

$= E_2 - 2E_1$

34. Answer (2)

From the given truth table output is low only when both inputs are high otherwise output is 1.

Hence the truth table is of NAND gate.

35. Answer (1)

Nuclear forces between two forces may also be repulsive for small separation between them.

SECTION - B

36. Answer (4)

N-type semiconductor is formed when pentavalent impurity is mixed to pure semiconductor.

37. Answer (2)

K.E. of photoelectrons is independent of intensity of incident photon and photoelectron have K.E. from Zero to maximum value.

38. Answer (1)

Momentum $P = \sqrt{2mK}$

According to de-Broglie $\lambda = \frac{h}{p}$

$$\lambda = \frac{h}{\sqrt{2mK}}$$

Hence, if K decrease then λ increase.

39. Answer (2)

Stopping potential for photoelectron is independent of intensity of incident photon.

40. Answer (2)

For stable nucleus,

$M < Zm_p + Nm_n$

41. Answer (3)

$A \to B + C$

$\Rightarrow$ for stable reaction,

$(B.E.)_{product} > (B.E.)_{Reactant}$

$E_b + E_c > E_a$

42. Answer (3)

Angular momentum in the n^{th} state is given by $\frac{nh}{2\pi}$

$\Rightarrow E_n = \frac{-13.6}{n^2} = -1.51$

$\Rightarrow n = 3$

Hence, $L = \frac{3h}{2\pi}$

43. Answer (4)

Energy $\propto \frac{1}{\text{wavelength}}$

Difference of energy is maximum for transition $n = 3$ to $n = 1$.

Hence minimum wavelength is for transition from $n = 3$ to $n = 1$.

44. Answer (1)

Young's modulus is the property of material.

Hence, it doesn't change with change in dimensions of wire.

45. Answer (1)

Least count = $\frac{\text{Pitch}}{\text{No. of division on circular scale}}$

$= \frac{1}{100}$ = 0.01 mm

46. Answer (3)

Minimum number of readings required are three.

47. Answer (2)

Energy of incident radiation.

$E = \phi + eV_0$

$= 5.3\text{ eV} + 4.2\text{ eV}$

$= 9.5\text{ eV}$

48. Answer (1)

Potential drop across $R = 15 - 10 = 5$ V

49. Answer (4)

A	*B*	*C*
1	0	0
0	1	0
1	1	1
0	0	0

The truth table corresponds to AND gate.

50. Answer (2)

$P = mv = 2 \times 100 = 200$

$$\lambda = \frac{h}{mv} = \frac{6.6\times10^{-34}}{200} = 3.3\times10^{-36}\text{ m}$$

CHEMISTRY

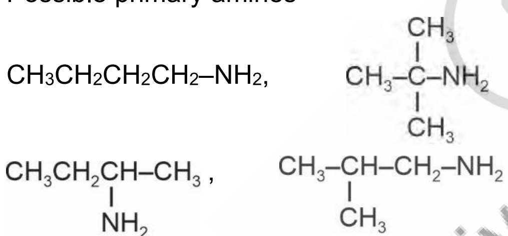
SECTION - A

51. Answer (2)

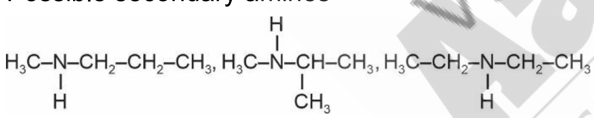
$-NO_2$ group decreases electron density on benzene ring and hence reactivity towards electrophilic substitution reaction decreases.

52. Answer (3)

Possible primary amines –



Possible secondary amines –



53. Answer (3)

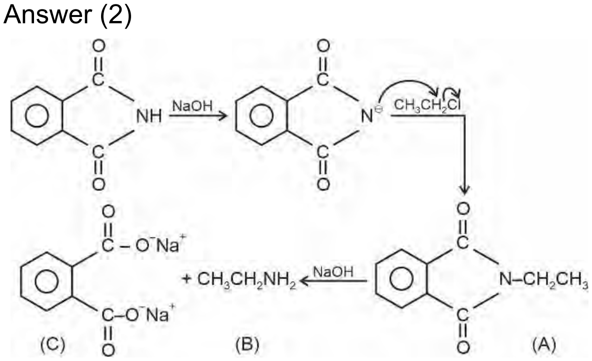
AgI is yellow and insoluble in NH_3

54. Answer (3)

$$PbCl_2 + KI \rightarrow \underset{\text{Yellow}}{PbI_2} + KCl$$

$$PbI_2 + \underset{\text{Excess}}{KI} \rightarrow K_2[PbI_4]$$

55. Answer (2)



56. Answer (4)

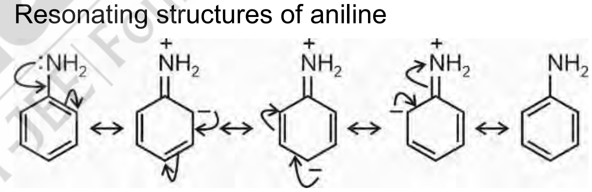
Globular proteins are spherical in shape *e.g.,* insulin. Myosin is the protein found in muscles.

57. Answer (2)

	BP (K)
$n-C_4H_9NH_2$	350.8
$(C_2H_5)_2NH$	329.3
$C_2H_5N(CH_3)_2$	310.5

58. Answer (2)

Resonating structures of aniline



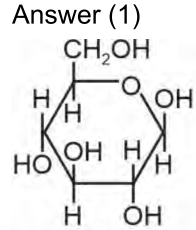
59. Answer (2)

$$K_2Cr_2O_7 + 14HCl \xrightarrow{\Delta} 2KCl + 2CrCl_3 + 7H_2O + 3Cl_2$$

60. Answer (1)

As hydrazine does not contain C it does not give Lassaigne's test.

61. Answer (1)



β-D-(+) Glucose

- It contains '5' chiral carbon and five –OH groups
- It contains pyranose ring.

62. Answer (2)

Maltase is used to convert maltose into glucose.

63. Answer (4)

Keratin and myosin are example of fibrous proteins.

64. Answer (1)

Vitamin $B_6 \rightarrow$ Pyridoxine

Vitamin C $\rightarrow$ Ascorbic acid

Vitamin $B_2 \rightarrow$ Riboflavin

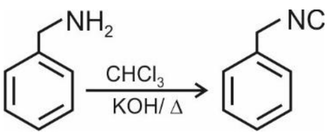
Vitamin $B_1 \rightarrow$ Thiamine

65. Answer (2)

Uracil is not present in DNA

Adenine, Thymine, Cytosine, Guanine are the bases present in DNA.

66. Answer (2)



67. Answer (4)

Tertiary amines do not react with Hinsberg reagent.

68. Answer (2)

Cellulose is a straight chain polysaccharide composed only of β-D-glucose units which are joined by glycosidic linkage between C_1 of one glucose unit and C_4 of the next glucose unit.

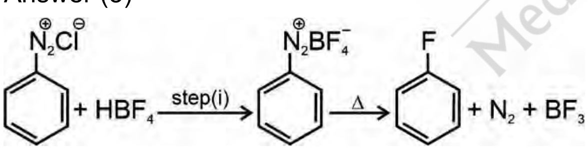
69. Answer (3)

$$NH_2-CH_2-CH=CH_2$$

Allyl amine

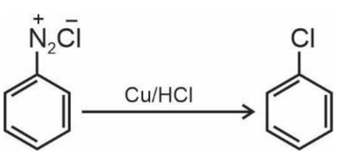
(Prop-2-en-1-amine)

70. Answer (3)

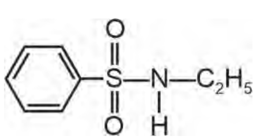


71. Answer (4)

Gattermann reaction



72. Answer (4)



N-Ethylbenzenesulphonamide

N-Ethylbenzenesulphonamide is soluble in alkali due to presence of acidic hydrogen atom.

Benzenesulphonyl chloride is used for the distinction of primary, secondary and tertiary amines.

73. Answer (3)

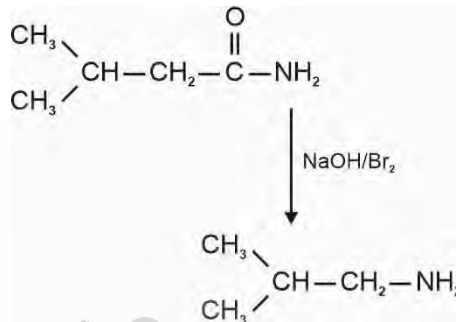
Alanine is a non-essential amino acid.

74. Answer (2)

Vitamin B and C are water soluble vitamins.

75. Answer (4)

Hoffmann Bromamide degradation



76. Answer (2)

α and β forms of glucose are anomers of each other due to the difference in configuration of hydroxyl group at C-1(anomeric carbon)

77. Answer (1)

$$\underset{\text{From sodium extract}}{S^{2-}} + [Fe(CN)_5NO]^{2-} \rightarrow \underset{\text{Violet}}{[Fe(CN)_5NOS]^{4-}}$$

78. Answer (2)

In acidic medium, Fe^{2+} ion is converted to Fe^{3+} ion.

$$5Fe^{2+}+MnO_4^-+8H^+ \rightarrow Mn^{2+}+4H_2O+5Fe^{2+}$$

Number of equivalents of $FeSO_4$ = number of equivalents of $KMnO_4$

79. Answer (4)

The compound which does not contain keto-methyl group or which does not give rise to keto-methyl group in reaction condition will not undergo haloform reaction.

80. Answer (4)

The Working pH range of phenolphthalein indicator is 8.2 to 10.2

81. Answer (1)

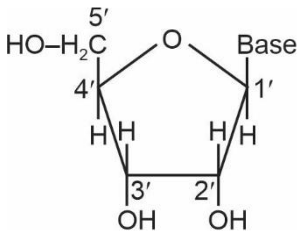
Group	Cation
Group-VI	Mg^{2+}
Group-V	Ba^{2+}, Sr^{2+} Ca^{2+}
Group-I	Pb^{2+}
Group-III	Al^{3+}, Fe^{3+}

82. Answer (2)

$Al_2O_3 \cdot xH_2O$ is a positively charged sol

83. Answer (1)

A unit formed by the attachment of a base to 1′ position of sugar is known as nucleoside.



Structure of a nucleoside

84. Answer (1)

CdS is yellow in colour.

85. Answer (1)

Starch in used in iodometric titration.

SECTION - B

86. Answer (4)

$$CH_3CH_2Br \xrightarrow{KCN} \underset{(A)}{CH_3CH_2CN} \xrightarrow{LiAlH_4} \underset{(B)}{CH_3CH_2CH_2NH_2} \xrightarrow[0°C]{HNO_2} \underset{(C)}{CH_3CH_2CH_2OH}$$

87. Answer (1)

Aniline does not undergo Friedel-Crafts reaction due to salt formation with $AlCl_3$.

88. Answer (1)

Nucleotides are joined together by phosphodiester linkage.

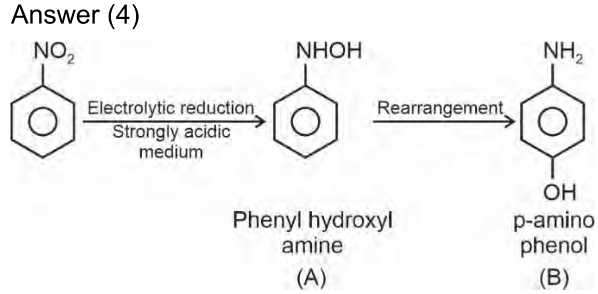
89. Answer (2)

Cheilosis is caused by deficiency of vitamin B_2.

90. Answer (2)

Starch is composed of α-D glucose units.

91. Answer (4)



92. Answer (2)

Ag^+, Hg_2^{2+} and Pb^{2+} form insoluble chlorides so they are grouped in group I.

93. Answer (1)

- COOH group gives effervescence of CO_2 with $NaHCO_3$.
- 2,4-DNP gives positive test with CHO.
- Hinsberg reagent is used for amines.

94. Answer (1)

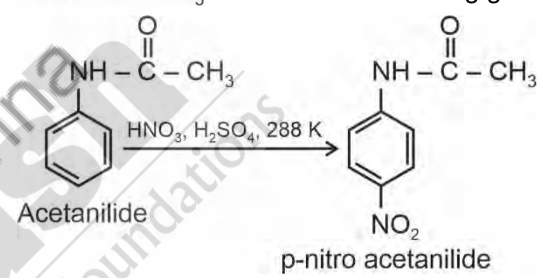
Aromatic amines are weaker bases than ammonia due to the electron withdrawing nature of the aryl group.

95. Answer (4)

In presence of acidic medium, ionisation of H_2S is suppressed so less number of S^{2-} ions are produced.

So only Cu^{2+} having low solubility product is precipitated.

96. Answer (1)

- $-NH-\overset{\overset{O}{||}}{C}-CH_3$ is an electron donating group
- 

97. Answer (3)

On heating, cobalt nitrate decomposes into CoO, which gives pink colour on reaction with MgO.

$$2Co(NO_3)_2 \xrightarrow{Heat} 2CoO + 4NO_2 + O_2$$

$$CoO + MgO \rightarrow \underset{Pink}{CoO \cdot MgO}$$

98. Answer (2)

Cations	**Colour**
Cu^{2+}	Blue
Co^{2+}	Blue, Red, violet, Pink
Ni^{2+}	Bright green
Mn^{2+}	Light pink

99. Answer (1)

$AlCl_3 + 3NaOH \rightarrow Al(OH)_3 + 3NaCl$

$$\underset{\text{White gilatinuous precipitate}}{Al(OH)_3} + NaOH \rightarrow \underset{\text{Sodium aluminate}}{NaAlO_2} + 2H_2O$$

Aluminium hydroxide is formed which is soluble in excess of sodium hydroxide solution due to formation of sodium aluminate.

100. Answer (4)

Ba^{2+} gives apple green colour to the flame test.

BOTANY

SECTION - A

101. Answer (2)

Some of the broadly utilitarian services that nature provides are oxygen, pollination, aesthetic pleasure and flood and erosion control.

102. Answer (3)

Thylacine extinct from Australia, Quagga from Africa, Steller's Sea cow from Russia and Dodo from Mauritius.

103. Answer (4)

Plants capture only 2-10% of PAR or 1-5% of incident solar radiation in synthesis of organic matter.

104. Answer (1)

Some small invertebrate animals that are called detritivores feed on detritus e.g. earthworms, termites. They bring about its fragmentation.

105. Answer (3)

Secondary succession starts in areas that somehow lost all the living organisms that existed there. Invading species depend on the condition of soil, availability of water, the environment and also the seeds or other propagules present. Climax is reached more quickly as compared to primary succession.

106. Answer (4)

Amount of all the inorganic substances present in an ecosystem per unit area at a given time is called standing state.

107. Answer (2)

Sacred groves are found in Aravali hills of Rajasthan.

Wildlife Safari parks are *Ex situ* conservation approaches.

Earth summit was held in Rio de Janeiro, 1992.

Hotspots have high degree of endemism.

108. Answer (4)

Among animals, insects are the most species rich taxonomic group making more than 70% of the total.

The ascending order will be Crustaceans < Molluscs < Insects.

109. Answer (3)

Tropical regions are less seasonal, relatively more constant and predictable. Warm and high humidity in tropical areas provide favourable conditions throughout year. Constant and favourable environment helps tropical organism to promote niche specialisation and lead to greater species diversity.

110. Answer (3)

Decomposition is largely an oxygen requiring process. The rate of decomposition is controlled by chemical composition of detritus and climatic factors.

111. Answer (2)

In Species-Area relationships given by Alexander von Humboldt $\log S = \log C + Z \log A$, S represents species richness.

112. Answer (3)

In the simplified model of phosphorus cycle, A represents detritus on which detritivores act.

X represents decomposition process, performed by microbes.

Y represent weathering of rocks which is not performed by detritivores.

113. Answer (4)

Hydra is an animal and it is a primary carnivore.

114. Answer (2)

River Dolphins are the national aquatic animal of India.

115. Answer (2)

Out of total quantity of global carbon, 71% carbon is found dissolved in oceans.

116. Answer (1)

Grass →	Grasshopper →	Frog →	Snake
1,00,000 J	10,000 J	1,000 J	100 J

117. Answer (4)

Nile perch, a large predator fish was introduced into Lake Victoria of east Africa. It is an alien species.

118. Answer (2)

All the biodiversity hotspots when put together covers less than 2% of earth's land area.

119. Answer (1)

The term biodiversity was popularised by Edward Wilson.

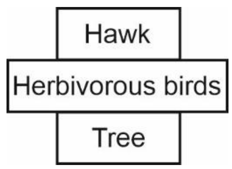
120. Answer (2)

Net primary productivity is gross primary productivity minus respiratory losses.

NPP = GPP - R

121. Answer (2)

Hawks are less in number as compared to herbivorous birds feeding on a tree, so the shape of pyramid is spindle shaped.



122. Answer (3)

The succession in aquatic habitat like pond is hydrosere. Pioneer community is formed by Phytoplanktons.

123. Answer (4)

For very large areas like entire continents, the slope of the line found to be much steeper (Z = 0.6 to 1.2)

124. Answer (1)

India has 14 Biosphere reserves and 90 National Parks.

125. Answer (2)

Rauwolfia vomitoria shows genetic diversity. Extinction of passenger pigeons is due to its over exploitation. Speciation is generally a function of time.

126. Answer (3)

Species at higher trophic level are more susceptible to extinction.

127. Answer (3)

Secondary productivity is the rate of formation of new organic matter by consumers. Biomass of different consumers are available as food for carnivores.

128. Answer (4)

In the process of photosynthesis, there will be fixation of CO_2 to produce sugars. Respiration, decomposition, combustion and burning of forests will release CO_2 and add to atmosphere.

129. Answer (3)

Annual net primary productivity of whole biosphere is approximately 170 billion tons. Energy at higher trophic level is always less than energy at a lower level.

130. Answer (2)

In Amazonian rain forest, number of birds species are 1,300, species of reptiles are 378 in number, amphibians are 427 and fishes species are 3,000 in number.

131. Answer (4)

Most of the land area of our country lies in tropics and has more than 1200 species of birds.

132. Answer (3)

The third stage of transitional communities in hydrarch succession is reed-swamp stage.

133. Answer (1)

Increase in biodiversity of an ecosystem contributes to increase in productivity.

134. Answer (2)

Strict protection of biodiversity hotspots could reduce the ongoing mass extinctions by almost 30%.

135. Answer (4)

Insects *i.e.*, invertebrates are the most diverse among animals. Angiosperms diversity is more than that of mosses.

SECTION - B

136. Answer (1)

IUCN maintain a Red Data Book or Red List which is a catalogue of threatened plant and animals facing risk of extinction.

137. Answer (3)

Animals which feed on green plants or plant products occupies second trophic level.

138. Answer (1)

Species diversity of amphibians is more in Western ghats than in Eastern ghats.

Bioprospecting is exploring molecular, genetic and species-level diversity for products of economic importance.

139. Answer (4)

The products of ecosystem processes which have environmental, aesthetic and indirect economic value are named as ecosystem services.

140. Answer (3)

Pyramid of energy is always upright, because the flow of the energy is unidirectional from producer to consumer level.

141. Answer (1)

India occupies 2.4% world's land area and 8.1% of global species diversity.

142. Answer (3)

The phenomenon of incorporation of nutrients in living microbes is called nutrient immobilisation.

143. Answer (2)

Ecosystem characteristics that changes during ecological succession are

- Total biomass increases
- Little diversity to high degree of diversity
- Aquatic or dry condition to mesic condition
- Increase in humus content of the soil

144. Answer (1)

Anthropogenic ecosystems do not possess self-regulatory mechanism.

145. Answer (2)

Tilman found that plots with more species showed less year-to-year variation in total biomass. He also showed that in his experiments, increased diversity contributed to higher productivity.

146. Answer (1)

Primary production in the ecosystem is measured either as weight (g m^{-2}) or energy (kcal m^{-2}).

147. Answer (4)

There are four major causes of biodiversity loss called 'The Evil Quartet'. These are

- Habitat loss and fragmentation
- Over exploitation
- Alien species invasion
- Co-extinction

148. Answer (3)

Outermost part of the biosphere reserve is transition zone, that allows human settlement.

149. Answer (1)

Amazon rain forest has highest biodiversity on earth.

150. Answer (4)

Secondary consumers or primary carnivores are animals which feed on herbivores, will form third trophic level.

Herbivores occupies second tropic level.

ZOOLOGY

SECTION - A

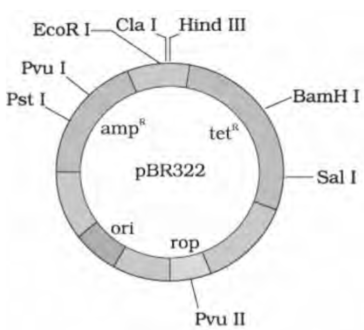
151. Answer (3)

Two enzymes responsible for restricting the growth of bacteriophages in *E. coli* are methylase and restriction endonucleases.

152. Answer (2)

Heat shock method is used to facilitate the uptake of DNA through the transient pores present in the bacterial cell wall.

153. Answer (4)



154. Answer (1)

The bacterial recognition sites for restriction enzymes are highly methylated, making them unrecognizable to the restriction enzyme.

155. Answer (2)

Many transgenic animals are designed to increase our understanding of how genes contribute to the development of diseases. These are specially made to serve as models for human diseases so that investigation of new treatments for diseases is made possible. Today transgenic models exist for many human diseases such as cancer, cystic fibrosis, rheumatoid arthritis and Alzheimer's.

156. Answer (4)

Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans (tobacco budworm, armyworm), coleopterans (beetles) and dipterans (flies, mosquitoes).

157. Answer (3)

Animals that have had their DNA manipulated to possess and express an extra (foreign) gene are known as transgenic animals.

Transgenic animals are used for:

(i) Studying normal physiology and development
(ii) Study of diseases
(iii) Obtaining useful biological products
(iv) Testing vaccine safety
(v) Investigation of a new treatment for auto-immune diseases
(vi) Performing chemical safety testing

158. Answer (3)

Plasmid DNA is double stranded, circular and does not carry any vital genes necessary for the cell. They are autonomously replicating extra-chromosomal DNA.

159. Answer (2)

PCR is used to amplify a given piece of DNA. Competency is the ability of cells to take up the alien DNA from its surrounding environment.

Transformation is a procedure through which a piece of DNA is introduced in a host bacterium.

160. Answer (3)

Golden rice contains precursor of vitamin-A *i.e*, β-carotene. The people with night blindness can get additional vitamin-A by consuming the golden rice.

161. Answer (4)

An ideal cloning vector is small in size. They contain *ori* site, selectable markers, and few (preferably single) restriction site for commonly used restriction enzymes.

162. Answer (3)

The Indian Government has set up organisations such as GEAC (Genetic Engineering Approval Committee), which makes decisions regarding the validity of GM research and the safety of introducing GM-organisms for public services.

163. Answer (1)

*Eco*R I cuts the DNA between bases G and A only when the sequence 5'GAATTC3' is present in the DNA in 5' $\rightarrow$ 3' direction.

Option (2) and option (4) are sequence of *Bam* HI and *Hin*d III.

164. Answer (2)

The number of DNA produced after 'n' number of PCR cycle = 2^n [where, 'n' = number of PCR cycles]

$\because 2^n = 2^5$ [n = 5]

= 2 × 2 × 2 × 2 × 2

= 32 DNA fragments will be obtained.

165. Answer (4)

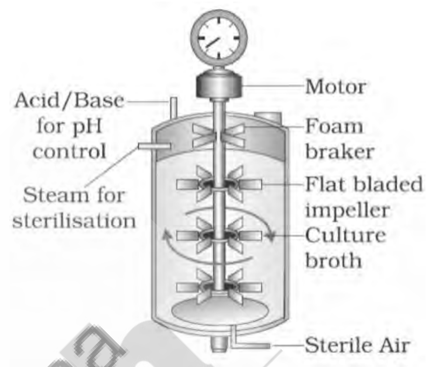
In PCR, extension of primers occurs from the 3′ end of growing strand.

Bacteriophages have the ability to replicate within bacterial cells independent of the control of chromosomal DNA. Bacteriophages because of their high number per cell, have very high copy numbers of their genome within the bacterial cells. Some plasmids may have only one or two copies per cell whereas others may have 15-100 copies per cell.

166. Answer (1)

Restriction enzymes cut the strand of DNA a little away from the centre of the palindrome sites, but between the same two bases on the opposite strands. This leaves single stranded portion at the ends. These are overhanging stretches, called sticky ends on each strand. These are named so because they form hydrogen bonds with their complementary cut counterparts.

167. Answer (3)



Sparger is a component of the sparged stirred-tank bioreactors.

168. Answer (4)

Genetically engineered insulin generally do not induce unwanted immunological responses. Also, its large scale production is possible.

Insulin used for diabetes was earlier extracted from pancreas of slaughtered cattle and pigs. Insulin from an animal source caused some patients to develop allergy or other types of reactions to the foreign protein.

169. Answer (3)

DNA appears as a collection of fine threads in the suspension after treatment with chilled ethanol which is then removed by spooling.

170. Answer (3)

In continuous culture system, used medium is drained out from one side while fresh medium is added from the other to maintain cells in their log phase. This type of culturing method produces a larger biomass leading to higher yields of desired proteins.

171. Answer (4)

Gene transfer with the help of vectors is called indirect gene transfer.

172. Answer (3)

Three critical research areas of biotechnology are:

(i) Providing the best catalyst in the form of improved organism usually a microbe or pure enzyme.

(ii) Creating optimal conditions through engineering for a catalyst to act, and

(iii) Downstream processing technologies to purify the protein/organic compound.

173. Answer (2)

In PCR, template DNA, *Taq* polymerase, primers and dNTPs are required.

dUTPs are not used for amplification of DNA as DNA lacks uracil in its structure.

174. Answer (4)

Eli Lilly used *E.coli* as the host to produce "Humulin" or recombinant insulin.

175. Answer (4)

The mature insulin lacks the C peptide chain. Whereas the pro-insulin contains all the three peptide chains (A, B and C).

176. Answer (3)

The proteins encoded by the gene *cryIAb* controls corn borer and that of *cryIAc* and *cryIIAb* controls the cotton bollworm.

177. Answer (2)

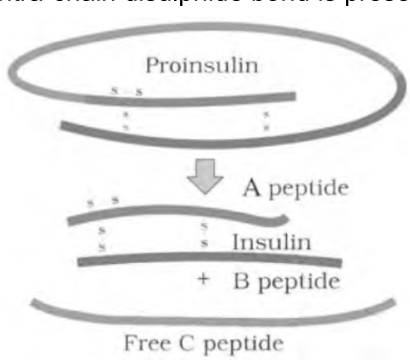
RNAi involves silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevent translation of the mRNA.

178. Answer (4)

Insulin is a protein hormone responsible for lowering the blood glucose levels. Insulin is never given orally as digestive juices will break it down before it reaches the blood stream.

179. Answer (4)

Intra-chain disulphide bonds are present within a single chain. In both pro-insulin and insulin, one intra-chain disulphide bond is present.



180. Answer (3)

PCR is an *in-vitro* gene cloning method used for amplification of DNA.

Serum and urine analysis are conventional methods of disease diagnosis.

181. Answer (1)

As a first step towards gene therapy, lymphocytes from the blood of the patient are grown up in a culture outside the body. A functional ADA cDNA (using a retroviral vector) is then introduced into these lymphocytes, which are subsequently returned to the patient. Lymphocytes are agranulocytes which constitute 20-25% of the total WBCs.

182. Answer (2)

B. thuringiensis forms protein crystals during a particular phase of their growth. These crystals contain a toxic insecticidal protein. This Bt toxin protein exists as inactive protoxins but once an insect ingests the inactive toxin, it is converted into an active form of toxin due to the alkaline pH of the gut which solubilise the crystals.

183. Answer (3)

A single stranded DNA or RNA, tagged with a radioactive molecule (probe) is allowed to hybridise to its complementary DNA in a clone of cells followed by its detection using autoradiography.

The clone having the mutated gene will hence not appear on the photographic film, because the probe will not have complementarity with the mutated gene.

184. Answer (4)

In 1983, Eli Lilly, an American company prepared two DNA sequences corresponding to A and B chains of human insulin and introduced them in plasmids.

Human protein (α-1-antitrypsin) is used to treat emphysema. Basmati rice is distinct for its unique aroma and flavour and 27 documented varieties of Basmati are grown in India.

185. Answer (2)

Biopiracy is the term used to refer to the use of bio-resources by multinational companies and other organisations without proper authorisation from the countries and people concerned without compensatory payment.

SECTION - B

186. Answer (2)

The cell wall of fungal cell is made up of chitin and hence, chitinase must be used for breaking the cell open to release its content. Along with it, ribonuclease and protease must be used which will digest the RNA and proteins respectively.

187. Answer (3)

Steps involved in genetically modifying an organism are:

- Identification and isolation of desired DNA
- Modification in a vector with same restriction enzyme as used for the desired DNA
- Introduction of desired DNA into the host cell
- Maintenance of introduced DNA in the host cell
- Transfer of desired DNA to the progeny of host

188. Answer (2)

*Hin*d II, isolated from *Haemophilus influenzae,* is the first restriction endonuclease which was isolated and characterised. Its recognition sequence is $\begin{matrix}5'GT(Py)(Pu)AC\ 3' \\ 3'CA(Pu)(Py)TG\ 5'\end{matrix}$ and it produces blunt ends.

EcoR I and *Bam*H I produces sticky ends.

Eco RV produces blunt ends.

189. Answer (2)

Agrobacterium tumefaciens, a pathogen of several dicot plants, is able to deliver a piece of DNA known as T-DNA to transform normal plant cells into a tumor and direct these tumor cells to produce the chemicals required by the pathogen.

190. Answer (3)

Restriction enzymes belong to a larger class of enzymes *i.e*. nucleases.

191. Answer (3)

The processes which include separation and purification of a product are collectively referred to as downstream processing. The product has to be formulated with suitable preservatives. Such formulation has to undergo through clinical trials as in case of drugs. Strict quality control testing for each product is also required.

192. Answer (4)

Bubbles dramatically increase the oxygen transfer area in the sparged stirred-tank bioreactors.

193. Answer (1)

It is essential that the explants, culture vessels, media and the instruments used for tissue culture be made free from microbes.

For this purpose, explants are treated with specific anti-microbial chemicals.

194. Answer (1)

In 1983, Eli Lilly, an American company prepared two DNA sequences corresponding to A and B chains of human insulin and introduced them in plasmids of *E. coli* to produce insulin chains. Chains A and B were produced separately, extracted and combined by creating disulfide bonds to form the human insulin (Humulin).

195. Answer (1)

PCR is used to amplify a given sample of DNA.

Western blotting is used to detect the presence of protein molecules.

Northern blotting is used to detect RNA and Southern blotting is used to detect DNA.

196. Answer (2)

In a chromosome, there is a specific DNA sequence called the ‘origin of replication’, which is responsible for initiating replication. Therefore, for the multiplication of any alien piece of DNA in an organism, it needs to be a part of a chromosome(s) which has a specific sequence known as ‘origin of replication’.

197. Answer (4)

Transformants are the ones which have pBR322 plasmid but not with the gene of interest whereas recombinants are the ones which have pBR322 plasmid with ligated gene of interest.

Therefore, recombinants which have foreign gene encoding enzyme, β-galactosidase, would appear blue in colour in the presence of its respective chromogenic substrate and will show sensitivity to ampicillin.

198. Answer (2)

DNA molecules are hydrophilic in nature.

199. Answer (2)

In 1963, the two enzymes responsible for restricting the growth of bacteriophage in *E.coli* were isolated. The first restriction endonuclease- *Hin*d II, whose functioning depends on a specific DNA nucleotide sequence was isolated and characterised five years later.

Each restriction endonuclease functions by ‘inspecting’ the length of a DNA sequence. Once it finds its specific recognition sequence, it binds to the DNA and cut each of the two strands of the double helix at specific points in their sugar - phosphate backbones.

200. Answer (2)

In a circular DNA, the number of restriction sites is equal to the number of DNA fragments obtained after restriction enzyme digestion whereas, in a linear DNA, the DNA fragments obtained are (1+ number of restriction sites).

Thus, DNA ‘X’ must be a linear DNA, whereas, DNA ‘Y’ must be a circular DNA.

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